

Intermediate Microeconomics — Week 3, Day 1

In-class Exercises: Duopoly — Bertrand, Cournot, Stackelberg

work in pairs • solutions at the end

Exercise 1 — Three ways to run one market [competency: solve each duopoly model]

A market has inverse demand $P = 100 - Q$, where $Q = q_1 + q_2$ is total output. Two identical firms each produce at constant marginal cost $c = 10$ and no fixed cost.

- (a) *Benchmarks.* Find the perfectly competitive price and quantity, and the monopoly price, quantity and profit.
 - (b) *Bertrand.* The firms compete by setting prices. State the equilibrium price, total quantity and each firm's profit, and explain in one sentence why the price ends up there.
 - (c) *Cournot.* The firms choose quantities simultaneously. Derive each firm's best-response function, solve for (q_1, q_2) , and report total Q , price P and each firm's profit.
 - (d) *Stackelberg.* Firm 1 chooses its quantity first, and firm 2 observes it before choosing. Solve by backward induction for q_1, q_2 , total Q , price P and each firm's profit.
 - (e) Rank the four structures (monopoly, Cournot, Stackelberg, Bertrand) by total quantity. Where does each duopoly sit relative to the two benchmarks?
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Exercise 2 — Welfare accounting [competency: surplus and deadweight loss]

Use the same market as Exercise 1 ($P = 100 - Q$, $c = 10$).

- (a) For the monopoly and for the Cournot outcomes, compute consumer surplus $CS = \frac{1}{2}Q(100 - P)$, industry profit, and total welfare $W = CS + \text{profit}$.
 - (b) The efficient (competitive) welfare is the whole triangle between demand and MC . Compute it, and then find the deadweight loss of the monopoly and of the Cournot duopoly.
 - (c) In one or two sentences: does adding a second (Cournot) competitor recover more or less than half of the welfare lost to monopoly? Give the number.
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Reconvene: be ready to state, for each structure, the *quantity*, the *price*, and one sentence on who gains and who loses relative to monopoly.